

Guidelines for Final Paper

1. You will write a paper as if you are submitting this to a journal and this is the first demonstration of your experiment. You will need to cite appropriate sources, format your figures properly, etc.
2. Write a normal journal style paper [in the PRL style](#), but with the following section headings:
 - a. Title, Author, Abstract
 - b. Introduction. Explain why this is relevant and its importance. Give a brief outline of the structure of your paper and result.
 - c. Theory section: Explain what you will be testing theoretically (for example, show the theoretical derivation for what you tested).
 - d. Experimental section: Explain the experimental apparatus and data taking protocol. You may cite sources that are appropriate for things like spontaneous parametric down conversion (SPDC), but the creation and verification of an entangled state from this process should be explained. You will need to make a figure (maybe two) of the experimental apparatus to explain how it works. Only include the parts that are important for a competent optical physicist to know about to reproduce your results (i.e. don't include lenses and mirrors that are not crucial or subtly important). **You must make these figures yourself** – they don't have to be fancy, but do need to be descriptive. Any figures that are even partially taken from others (even with citations) will constitute academic dishonesty and will be treated as such.
 - e. Results section: Present and explain experimental data. I'd like to see at least 1 plot of some sort so the reader can understand your results and their uncertainty (if appropriate). You must clearly explain how you are dealing with experimental uncertainties so the reader can clearly understand your rationale.
 - f. Conclusion: Based on data and your error estimate, state the outcome of your experiment (e.g. violated Bell's inequality by 3 standard deviations, etc.). Interpret your results. If your results don't agree perfectly with theory (spoiler: they won't be) give some reasons, with supporting details, on what may have limited them.
 - g. Summary: Recap what you have demonstrated and what this may mean for the future.
3. **Due by 11:59pm on June 12 via Gradescope. No late submissions!**

Grading scale:

The requested parts of the manuscript will be assigned points as follows:

- a. Title, Author, Abstract.
 - i. 10 pts. Mostly this is assigned based on abstract containing necessary info.

- b. Introduction. Explain why this is relevant and its importance. Give a brief outline of the structure of your paper and result.
 - i. 8 pts. Relevance/Importance
 - ii. 2 pts. Brief outline
- c. Theory section: Explain what you will be testing theoretically
 - i. 25 pts. Give a theoretical derivation of the physics effect you will be testing/demonstrating. Give predictions on what you should measure.
- d. Experimental section: Explain the experimental apparatus
 - i. 10 pts: explain the experimental apparatus
 - ii. 10 pts: original figure (or figures) showing relevant parts
 - iii. 5 pts: experimental protocol
- e. Results section: Present experimental data
 - i. 15 pts. Description of experimental data (including a figure)
 - ii. 10 pts. Discussion of discrepancies between your data and theory and/or uncertainties.
- f. Conclusion: Based on data and your error estimate, state the outcome of your experiment (e.g. violated Bell's inequality by 3 standard deviations, etc.) and interpret.
 - i. 10 pts
- g. Summary
 - i. 5 pts.

Feedback: I will do my best to give you feedback on a draft of your paper if you send it to us by Friday, June 5 (or earlier!).